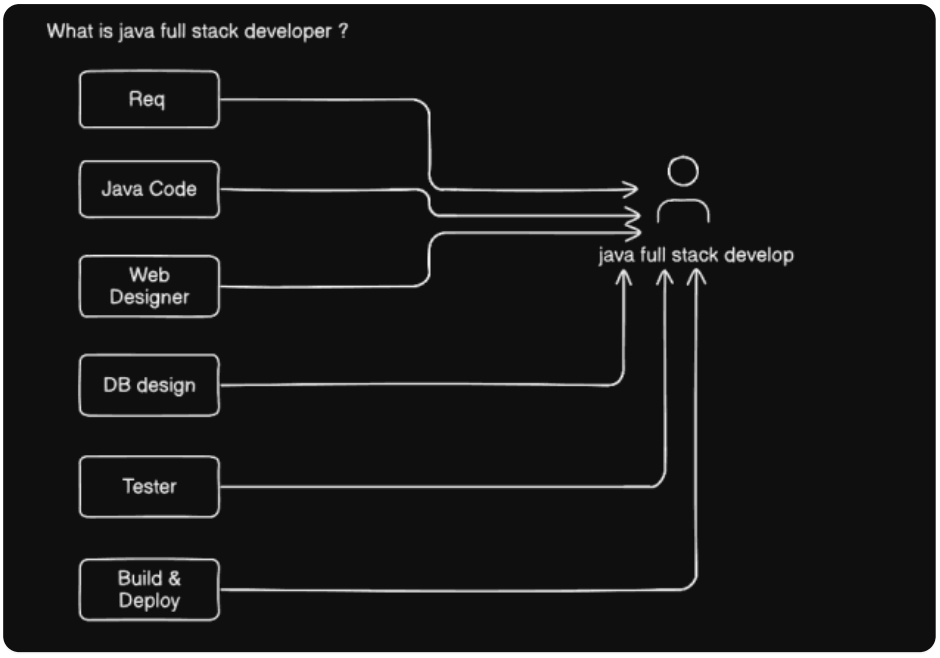
Introduction Java

07 June 2026 09:05

What is java developer ?



Java Application Architectures:

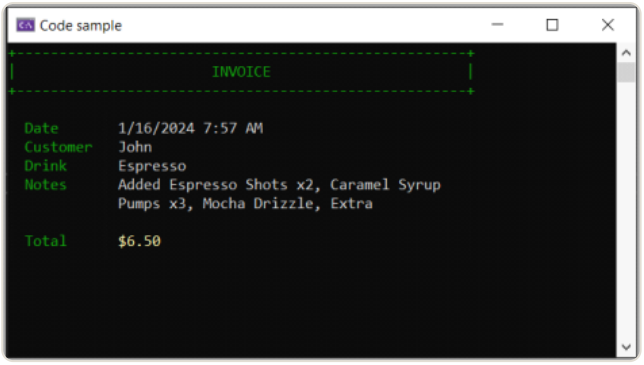
01 Standalone Applications

> **Standalone Applications :** An application that is installed on single isolated machine and executed it's action entirely within same computer /machine

a . **Core Technology stack :** Core java (J2SE - java 2 standard edition)

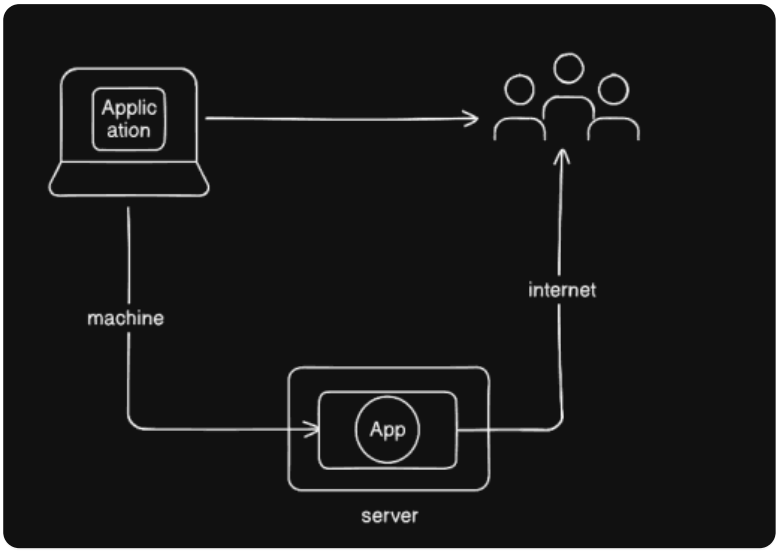
Developer perspective :

1 **No HTML:**

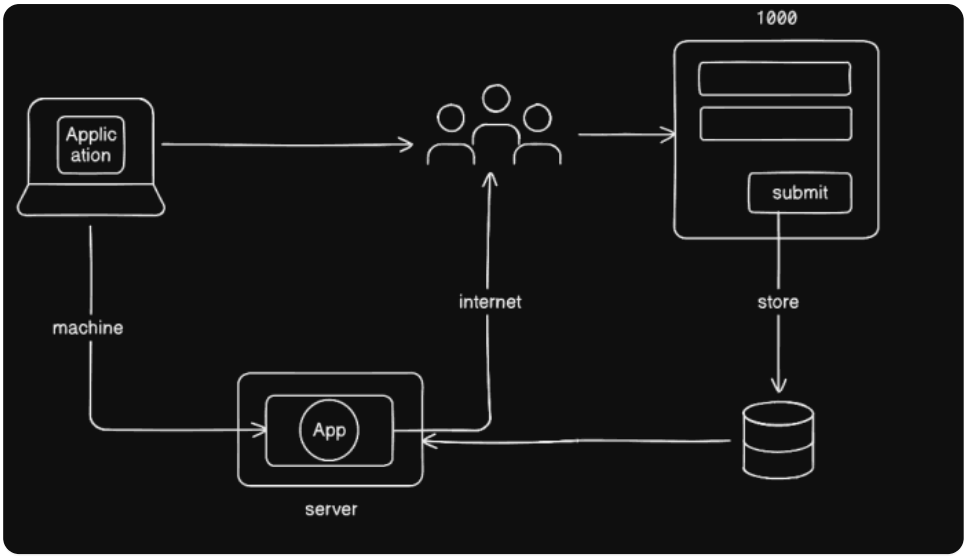


2 **No Server environment:** isko run karne ke liy kisi external machine ki jarurat nhi ha

? What is server ?



3 **NO Database storage:**



02 Web Application

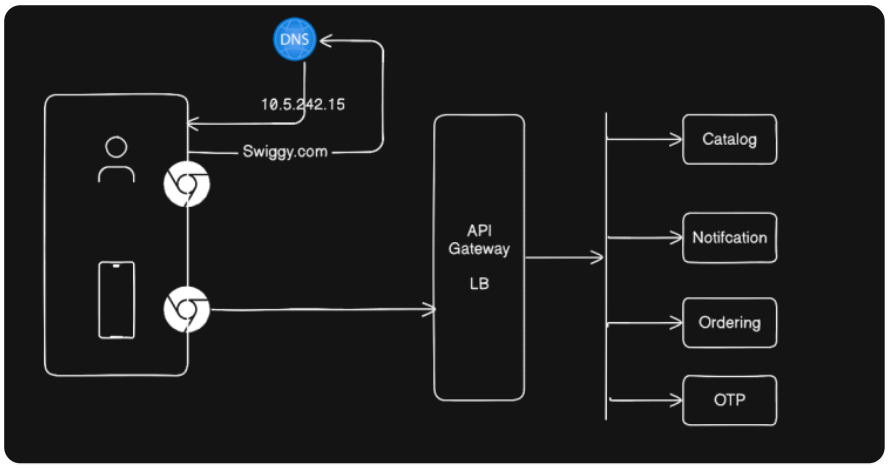
> An application that executes with Web environment or internet environment is known as web application

Tech Stack:

`1. JDBC` `2. Servlet` `3. JSP`

03 Enterprise Application

> An application that execute within a distributes environment and relies heavily on advance infra services like **security**, **load balancer**, **clustering** Is knows EA or EDP



Core Concepts of Software Development:

1 Programming language

A language that provides syntax rule, structure and concepts

It tells system how to allocate memory, manage data types, evaluate condition, execute conditions

Eg:	lang=vocabulary of English
Core java :	provide basic syntax, classes, interfaces

2 Technology

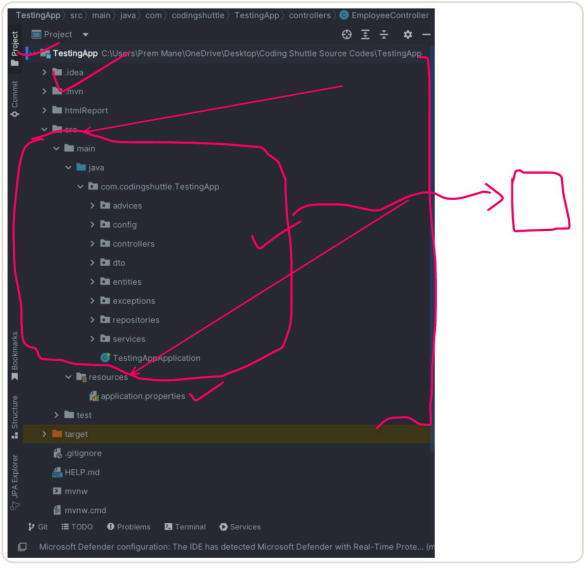
The process of converting theoretical knowledge and language concept into Realtime functionality or application development is known as **technology**

Technology act as bridge/specification that utilize a language to solve specific problem(like web communication, database connectivity etc)

3 Framework

Framework: is a semi- built, ready-constructed architectural structure available for application development.

It provides pre-written (boilerplate code), standard design patterns and lib allowing developers to focus on business login instead of setting up everything scratch.



Fundamentals of programming & Code Execution lifecycle

Har ek developer ko pata hona chahiye ki backend par code likhne se lekar uske run hoke tak system ke ander kya kya steps hote ha

- **Program:** set of instruction given to machine to perform a specific task
- **Programming:** The overall process used to construct program. It specifically means converting Analysis and design into readable source code.
- **Programmer:** The person who writes the program/source code.

Code Lifecycle stages:

Jab ek programmer source code likhta ha to use respective language extension k sath save krta ha

eg: `Test.c` , `Test.cpp` , `Test.py`
Eg: `Test.java`



STAGE A

compilation process

Process of checking whether the constructed program follow the strict syntax rules and grammar of that specific programming language is known as **compilation process**.

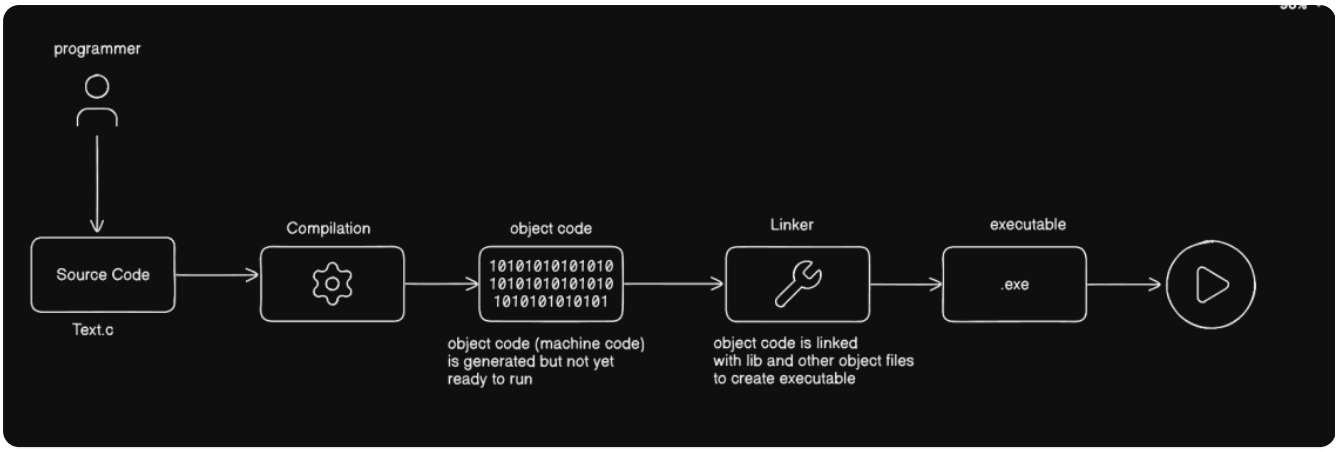
- **The Compiler:** lang like C, C++ and java use dedicated s/w component called **compiler** to scan and verify the code you have written.

STAGE B

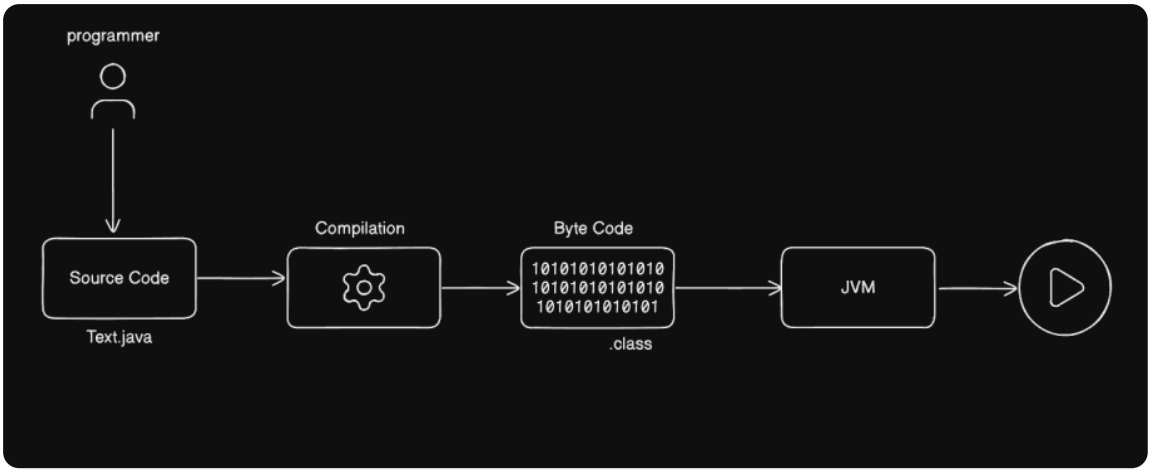
Execution Process:

The process of running the compiled code to verify the required and expected output generated or not is known as **execution process**.

execution steps of C lang:



Java Code Execution flow:



Objective code Vs Byte Code:

